

SCREEN-SHOOTING RESILIENT WATERMARKING BASED ON LONG-RANGE MODELING AND HIGH-FREQUENCY ENHANCEMENT

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ABSTRACT

Screen-shooting has become a major threat to image watermarking in the era of mobile devices. Most existing approaches focus on increasingly sophisticated distortion simulation during training, yet still adopt short-range network backbones. Such local-scale feature fusion is insufficient to model cross-region dependencies under nonlinear screen-shooting degradations, leading to unreliable decoding. In this paper, we propose WNet, an end-to-end watermarking framework based on a cascaded dual-UNet architecture. WNet establishes long-range dependencies to effectively capture global distortion patterns and enhance resilience against complex real-world degradations. To mitigate the attenuation of fine details introduced by long-range modeling, we design a Laplacian-pyramid-based high-frequency enhancement channel attention module to explicitly strengthen high-frequency structures. In addition, a texture edge guided loss is constructed to align watermark information with invisible texture-rich regions. Experimental results demonstrate that the proposed scheme significantly outperforms existing state-of-the-art methods. On average, our method achieves 1.28% higher bit accuracy and a 0.52 dB gain in terms of PSNR compared with DoBMark.

Index Terms— Screen-shooting, Watermarking, Long-range modeling, High-frequency enhancement, Texture edge guided loss

1. INTRODUCTION

Information security issues have received increasing attention with the development of digital technology. Digital watermarking technology is one of the effective solutions to these issues and has been widely applied in copyright protection and information tracking [1, 2]. However, screen-shooting has become an increasingly prevalent method of image theft,

leading to distortions such as geometric deformations, lighting variations, blurring, and moiré patterns [3, 4, 5]. These degradations destroy the structural and frequency characteristics of images, making watermark extraction unreliable [6, 7].

Traditional screen-shooting resilient watermarking algorithms typically rely on qualitative analysis of distortion during the shooting process and adopt fixed embedding strategies to enhance robustness [8, 9]. While these methods offer simplicity and computational efficiency, they lack adaptability to uncontrolled shooting conditions, leading to frequent failures in watermark recovery.

Deep learning-based watermarking algorithms allow the network to dynamically adapt to various distortions and effectively exploit the fine-grained features and spatial structures of the host image for improving model performance [10, 11, 12, 13]. Tancik et al. proposed an end-to-end print-shooting resilient watermarking neural network called StegaStamp [10]. Based on StegaStamp, Guo et al. integrated a series of mathematical operations in a distortion network to simulate multiple shooting methods [11]. By training with style transfer techniques, Gao et al. developed a realistic and credible comprehensive distortion layer [12]. Fang et al. proposed a distortion layer dubbed PIMoG, which can simulate common screen-shooting distortions in a differentiable way [13]. Wengrowski et al. introduced a real-time system that embeds and recovers hidden information in screen-displayed videos captured by handheld cameras [14]. Although these methods enhance robustness to some extent, they primarily rely on designing distortion layers and employing short-range modeling architectures, such as UNet, for feature extraction and reconstruction. These structures exhibit limitations when it comes to capturing cross-region dependencies and handling large-scale geometric and nonlinear distortions. Specifically, the UNet encoder can fuse features across different levels, but this fusion remains limited to local scales, which causes the degradation of watermark information in large-scale screen-shooting distortion scenarios.

To address the above concerns, we propose a long-range WNet-based framework that incorporates a Laplacian pyra-

* is the corresponding author. This work was supported by NSFC (Grant No. T2541025), Natural Science Foundation of Guangdong Province (No. 2025A1515012740), and Guangdong Key Laboratory of Information Security Technology (No. 2023B1212060026).

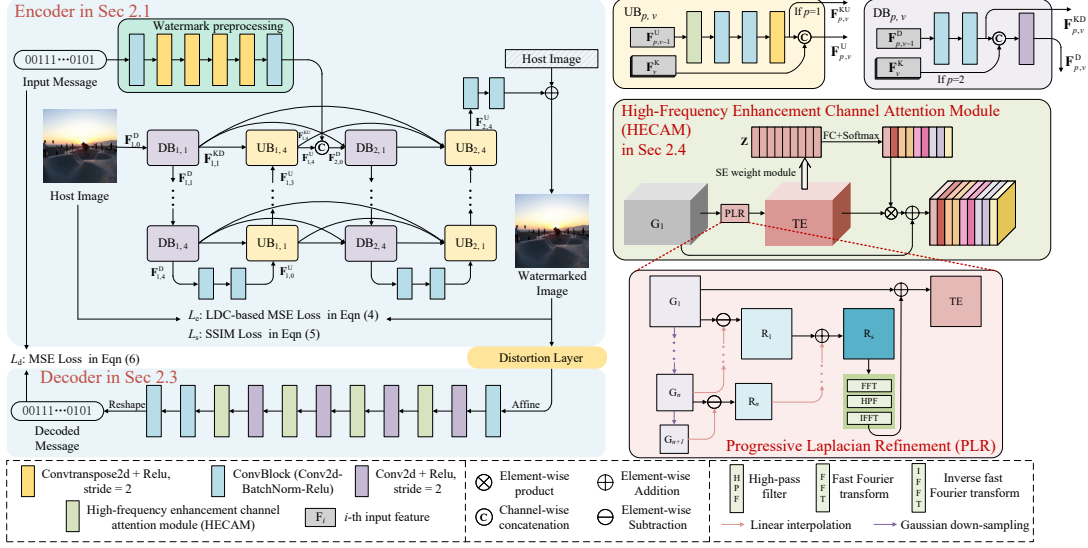


Fig. 1. Overview of the proposed screen-shooting resilient watermarking framework.

mid attention mechanism and a texture edge guided loss to improve both robustness and invisibility in screen-shooting scenarios. Our contributions are as follows:

1) We propose a cascaded dual-UNet architecture, WNet, which establishes long-range dependencies through a two-stage UNet pipeline. This model first extracts multi-scale host priors and then integrates the watermark with these priors, while dense cross-stage skip connections promote cross-region feature propagation for more robust embedding.

2) We design a high-frequency enhancement channel attention module, HECAM. By decomposing multi-scale features with a Laplacian pyramid and re-weighting high-frequency responses via channel attention, HECAM preserves local structural details during feature fusion and provides stable carriers for watermark embedding.

3) We introduce a texture edge guided mean squared error loss to guide the distribution of watermark information toward the texture edge region of the image, improving the invisibility of the watermarked images.

2. PROPOSED MODEL

The proposed watermarking framework includes three parts: encoder, decoder, and distortion layer, as shown in Fig. 1.

2.1. Encoder

To capture cross-region dependencies that are essential for resisting large-scale screen-shooting distortions, we propose a cascaded dual-UNet encoder, termed WNet. The two-stage architecture enables long-range feature propagation, allowing the network to integrate global context for robust watermark embedding. It processes the host image $\mathbf{I}_0$ in two sequential stages. The first stage extracts multi-scale visual features, and the second stage integrates the watermark information with these features for robust embedding. The network utilizes dense skip connections to ensure comprehensive fusion of contextual information across and within stages. Each

stage consists of a series of down-sampling blocks $\text{DB}_{p,v}$ and up-sampling blocks $\text{UB}_{p,v}$, where the stage index $p \in \{1, 2\}$ and the block index $v \in \{1, 2, 3, 4\}$. In particular, the DB blocks in Stage-1 do not incorporate skip connections as inputs, and the UB blocks in Stage-2 do not generate skip features, since they are located at the start and end of the network path, respectively. In the first stage, the network takes $\mathbf{I}_0$ as input. $\text{DB}_{1,v}$ performs convolution followed by down-sampling, generating a down-sampled feature map $\mathbf{F}_{1,v}^D$ for the next block $\text{DB}_{1,v+1}$ and a skip feature $\mathbf{F}_{1,v}^{\text{KD}}$. Subsequently, $\text{UB}_{1,v}$ performs feature reconstruction. At the beginning of $\text{UB}_{p,v}$, HECAM is integrated. It decomposes the input features using a Laplacian pyramid, extracts multi-scale high-frequency components, and assigns higher weights to these texture-rich features through a channel attention mechanism. The enhanced feature is up-sampled and concatenated with the skip features. In this stage, $\text{UB}_{1,v}$ output a skip feature map $\mathbf{F}_{1,v}^{\text{KU}}$ before the concatenation operation. The final output of $\text{UB}_{1,4}$ is denoted as $\mathbf{F}_{1,4}^U$, which contains the host image's multi-scale prior information. The second stage is responsible for embedding the watermark message $\mathbf{M}$. $\mathbf{M}$ is first reshaped and up-sampled through a small network to match the spatial resolution of $\mathbf{F}_{1,4}^U$. This processed watermark is then concatenated with $\mathbf{F}_{1,4}^U$ along the channel dimension. The combined tensor $\mathbf{F}_{2,0}^D$ serves as the input to the first down-sampling block $\text{DB}_{2,1}$. The down-sampling and up-sampling paths in Stage-2 operate identically to Stage-1 in terms of structure and skip connection mode. The final output of this stage, i.e., $\mathbf{F}_{2,4}^U$ is processed through a final convolution layer and added to $\mathbf{I}_0$ to generate the watermarked image $\mathbf{I}_e$.

2.2. Distortion layer

To verify the performance gain brought by the network framework, the same distortion layer as in PIMoG [13] is adopted to

simulate real-world capture distortions. In addition, six common distortions are introduced to enhance the model's robustness in electronic transmission scenarios. These distortions include resizing, JPEG compression, Gaussian blur, salt-and-pepper noise, dropout, and cropout.

2.3. Decoder

In the decoding process, four-point perspective transformation is employed to achieve precise correction by rectifying the quadrilateral boundaries or projection relationships of the captured image. Each down-sampling operation doubles the number of feature channels to compensate for resolution loss. HECAM improves watermark extraction by assigning higher weights to feature maps containing texture-rich regions. The downstream convolutions reduce the feature map to a single channel, and the final output is reshaped to ensure dimensional alignment with the input watermark message.

2.4. HECAM

Screen-shooting distortions often attenuate high-frequency details of the watermarked image, which are critical for watermark embedding. To preserve these details in long-range modeling, we assign higher weights to high-frequency features via HECAM. This ensures that texture-rich regions, which are more resilient to geometric and photometric distortions, retain watermark information. Inspired by Hu et al. [15] and Zhang et al. [16], a high-frequency enhancement channel attention module, i.e., HECAM, is designed to capture the high-frequency features of the input. At the core of HECAM lies the progressive Laplacian refinement (PLR), which refines the traditional Laplacian decomposition into a multi-scale, frequency-aware representation. Unlike manual filters or global frequency transformations, PLR can progressively separate high-frequency components while preserving spatial correlations. By leveraging the hierarchical structure of the Laplacian pyramid, the network can focus on high-frequency components, enhancing its ability to emphasize fine-grained texture information. The structure of HECAM is shown in Fig. 1, and the detailed calculation process is summarized in Algorithm 1. Compared to hand-crafted DCT and gradient priors, the pyramid-based structure offers better alignment with convolutional feature hierarchies and provides both localization and frequency discrimination. A high-pass filter is employed to further suppress low-frequency noise during the progressive refinement process of the Laplacian pyramid, and to highlight the high-frequency texture features. The filter $HP(\cdot)$ is formulated as:

$$HP(u, v) = 1 - \exp\left(-\frac{D(u, v)^2}{2\sigma^2}\right) \quad (1)$$

$$D(u, v) = \sqrt{(u - u_0)^2 + (v - v_0)^2} \quad (2)$$

where $D(u, v)$ represents the Euclidean distance from (u, v) to the spectrum center (u_0, v_0) , with $(u_0, v_0) = (\frac{H}{2}, \frac{W}{2})$ for an $H \times W$ spectrum. σ controls the cutoff frequency and determines the sharpness of the filter transition. After multiple

tests, we set σ to 0.1 to suppress low-frequency noise while preserving the high-frequency structural information of the feature maps.

Algorithm 1: Proposed HECAM

Input: Input feature map $\mathbf{X}$, i.e., $\mathbf{G}_1$, the number of channels C

- 1 The input $\mathbf{X}$ is passed through PLR to extract fine-grained high-frequency feature $\mathbf{TE}$;
- 2 $\mathbf{G}_{i+1} = \text{downsample}(\mathbf{G}_i)$, $i \in \{1, 2, \dots, n\}$;
- 3 $\mathbf{R}_s = \sum_{i=1}^n (\mathbf{G}_i - \text{upsample}(\mathbf{G}_{i+1}))$;
- 4 **for** $c=0$ to $C-1$ **do**
- 5 $\mathbf{F}_F = \text{FFT2D}(\mathbf{R}_s[c, :, :])$;
- 6 $(H, W) = \text{SpatialSize}(\mathbf{F}_F)$;
- 7 **for** $u=0$ to $H-1$, $v=0$ to $W-1$ **do**
- 8 $\mathbf{F}_T(u, v) = \mathbf{F}_F(u, v) \odot HP(u, v)$;
- 9 **end**
- 10 $\mathbf{TE}[c, :, :] = \text{IFFT2D}(\mathbf{F}_T) \cdot \text{real}$;
- 11 **end**
- 12 Extract channel attention weights:
- 13 $\mathbf{Z} = \text{SE}(\mathbf{TE})$; where $\text{SE}(\cdot)$ denotes the squeeze-and-excitation network in [15].
- 14 Apply Softmax function and fully connected layer:
 $\mathbf{F} = \text{Softmax}(\text{FC}(\mathbf{Z})) \odot \mathbf{TE} + \mathbf{X}$.
- 15 **return** $\mathbf{F}$

2.5. Loss function

To generate watermarked image $\mathbf{I}_e$ with high visual quality and decoded message $\mathbf{M}_d$ with high accuracy, the overall optimization object is:

$$L_t = L_e + \lambda_1 L_s + \lambda_2 L_d \quad (3)$$

where λ_1 and λ_2 are the weight coefficients to balance the contributions of the respective loss components. Through multiple comparative experiments, they are determined to be 0.1 and 4, respectively.

2.5.1. Encoding loss L_e & L_s

We formulate L_e as the sum of a reconstruction term and a texture edge guidance term. The reconstruction term constrains the distortion between $\mathbf{I}_o$ and $\mathbf{I}_e$. The guidance term aligns the embedding watermark residual $\mathbf{I}_r$ with the texture edge mask extracted from $\mathbf{I}_o$ by the pre-trained lightweight dense CNN (LDC) [17], thereby encouraging watermark modifications to concentrate on texture edge regions. Here, $\mathbf{I}_r = \mathbf{I}_e - \mathbf{I}_o$. L_e is denoted as:

$$L_e = \text{MSE}(\mathbf{I}_o, \mathbf{I}_e) + \text{MSE}(\tau \cdot \text{LDC}(\mathbf{I}_o), \mathbf{I}_r) \quad (4)$$

where τ controls the strength of the texture edge mask and defaults to 1. Moreover, structural similarity (SSIM) is applied to measure the distortion of the watermarked image at the perceptual level in terms of brightness, contrast, and structure. The structural loss L_s can be defined as:

$$L_s = 1 - \text{SSIM}(\mathbf{I}_o, \mathbf{I}_e) \quad (5)$$

2.5.2. Decoding loss L_d

The decoder aims to minimize the difference between $\mathbf{M}_d$ and the original message $\mathbf{M}$. Therefore, the decoding loss L_d is defined as

$$L_d = \text{MSE}(\mathbf{M}, \mathbf{M}_d) \quad (6)$$

3. EXPERIMENTAL RESULTS

3.1. Experimental Setup

To train the proposed network, 10,000 images are randomly selected from the COCO dataset as the training dataset. All images are resized to 128×128×3 pixels, and the length of the watermark message is 64. The framework is implemented by PyTorch and executed on NVIDIA GeForce RTX 4060 Ti. Parameters optimization of the network is implemented by the Adam optimizer with a learning rate of 1e-3 and betas of (0.9, 0.999). The batch size is 16 and the training epoch is 100. The depth of the PLR defaults to 2. The mobile shooting device and the image monitor use IQOO Neo 5S and Lenovo Display HDR, respectively. Peak signal-to-noise ratio (PSNR), SSIM, and learned perceptual image patch similarity (LPIPS) are selected as the evaluation metrics. Bit accuracy (BA) is used to evaluate robustness by calculating the accuracy of decoded messages. The proposed scheme is compared with the benchmark Stega [10] and LFM [14], the state-of-the-art (SOTA) schemes DoBMark [11], PIMoG [13] and ABStyle [12]. Particularly, we clarify that, in the tables, we reported the results of PIMoG and ABStyle from the original papers, as indicated by the symbol *, since they apply a watermark message with a length of 30 and we could not reliably reproduce the results. In the following tables and figures, the suffixes _30 and _64 denote embedding capacities of 30 bits and 64 bits, respectively.

3.2. Visual quality

The invisibility of different watermarking schemes is compared in Table 1. The PSNR of the proposed scheme is lower than that of ABStyle. However, due to the distribution of watermark information in texture-rich high-frequency regions, the proposed scheme achieves the highest SSIM, indicating optimal visual quality.

Table 1. Invisibility assessment of different schemes (_64 and _30 denote the embedded message length)

Scheme	PSNR (dB)	SSIM	LPIPS
Stega_64 [10]	28.35	0.9026	0.051
DoBMark_64 [11]	34.77	0.9673	0.017
LFM_64 [14]	32.92	0.9566	0.024
Proposed_64	35.29	0.9868	0.006
ABStyle_30 [12]*	38.20	0.9840	-
PIMoG_30 [13]*	36.21	0.9850	-
Proposed_30	36.69	0.9912	0.004

3.3. Performance evaluation with screen-shooting

3.3.1. Shooting distance

The shooting distance ranges from 20 cm to 60 cm, with an interval of 10 cm for each capture. As shown in Table 2, the proposed watermarking scheme exhibits superior decoding efficiency within the tested distance range. Notably, the

robustness of some schemes improves when the shooting distance increases from 40 cm to 50 cm. A longer distance reduces image details but also weakens sensor–screen sampling interference, alleviating the impact of moiré patterns.

Table 2. Decoding BA values at different shooting distances

Scheme	20 cm	30 cm	40 cm	50 cm	60 cm
Stega_64 [10]	97.2%	96.7%	96.4%	95.2%	95.3%
DoBMark_64 [11]	98.5%	97.8%	98.3%	98.3%	98.0%
LFM_64 [14]	92.9%	92.7%	91.5%	92.5%	91.0%
Proposed_64	99.5%	99.0%	98.5%	98.6%	98.2%
ABStyle_30 [12]*	98.4%	98.0%	98.2%	-	-
PIMoG_30 [13]*	99.4%	98.7%	98.0%	97.3%	98.7%
Proposed_30	99.8%	99.5%	99.5%	99.7%	99.4%

3.3.2. Shooting angle

The experiment is conducted by positioning the mobile device at different angles relative to the center of the display image. The shooting angles are 20°, 30°, and 40° from left to right and from top to bottom. Table 3 exhibits that Proposed_30 maintains BA above 99% across different shooting angles. This demonstrates that long-range modeling combined with high-frequency enhancement enables the network to preserve critical details across large geometric variations, thereby maintaining high decoding accuracy.

Table 3. Decoding BA values at different shooting angles

Scheme	L 40° / T 40°	L 30° / T 30°	L 20° / T 20°
Stega_64 [10]	88.0% / 96.7%	95.7% / 96.2%	96.2% / 95.0%
DoBMark_64 [11]	92.7% / 97.5%	97.5% / 97.7%	98.1% / 98.2%
LFM_64 [14]	84.8% / 91.5%	89.2% / 89.6%	91.0% / 90.2%
Proposed_64	96.8% / 98.6%	98.6% / 98.6%	97.7% / 98.8%
ABStyle_30 [12]*	-	97.1% / -	-
PIMoG_30 [13]*	97.1% / 98.7%	98.1% / 98.7%	98.7% / 99.2%
Proposed_30	99.0% / 99.0%	99.3% / 99.4%	99.2% / 99.6%
Scheme	R 40° / B 40°	R 30° / B 30°	R 20° / B 20°
Stega_64 [10]	96.0% / 95.7%	96.7% / 95.2%	96.7% / 96.0%
DoBMark_64 [11]	95.8% / 95.5%	96.5% / 96.8%	97.9% / 97.3%
LFM_64 [14]	90.6% / 91.0%	90.0% / 91.7%	89.4% / 91.4%
Proposed_64	99.0% / 98.0%	99.0% / 98.3%	99.2% / 97.7%
ABStyle_30 [12]*	-	98.3% / -	-
PIMoG_30 [13]*	99.4% / 99.2%	98.9% / 98.7%	99.0% / 99.0%
Proposed_30	99.3% / 99.3%	99.8% / 99.5%	99.7% / 99.1%

3.3.3. Shooting device

The above experiments are conducted with IQOO Neo 5S and Lenovo Display HDR. To verify the adaptability of the proposed scheme, two devices, Honor 90 GT and iPhone 15 Pro Max, and one monitor, Sanc, are introduced, and the model is tested under different combinations. The shooting angle and distance are fixed at 0 and 50 cm. It can be concluded from Table 4 that the proposed watermarking scheme achieved a higher BA than Stega_64 and LFM_64, with an increase of approximately 2.6% and 6%, respectively.

Table 4. Decoding BA values under different devices

Display	Camera	Stega_64 [10]	DoBMark_64 [11]	LFM_64 [14]	Proposed_64
Lenovo	IQOO	96.2%	98.6%	92.3%	98.9%
	Honor	95.7%	98.3%	91.8%	98.3%
	Iphone	96.0%	97.9%	92.1%	98.7%
Sanc	IQOO	95.9%	98.5%	92.5%	98.6%
	Honor	95.4%	97.9%	91.6%	97.8%
	Iphone	95.7%	97.9%	92.2%	98.3%

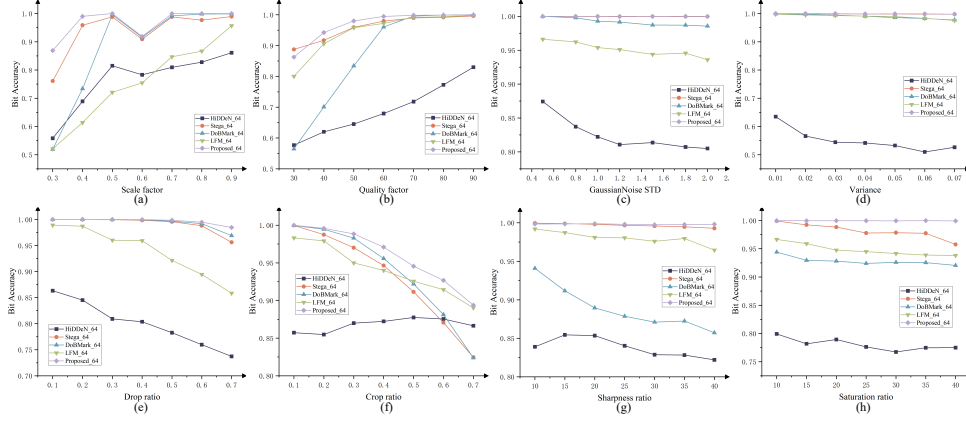


Fig. 2. Accuracy under different distortions in electronic transmission channels. (a) Resizing; (b) JPEG compression; (c) Gaussian Blur; (d) Salt and Pepper noise; (e) Dropout; (f) Cropout; (g) Adjust sharpness; (h) Adjust saturation.

3.4. Performance evaluation in electronic transmission

The model performance in electronic transmission scenarios is evaluated by testing on six distortions included in training and two additional distortions not used in training: sharpness adjustment and saturation adjustment. All compared models are trained with the same distortions to ensure fairness. As shown in Fig. 2, the proposed method achieves superior decoding efficiency under most distortions. For JPEG compression, when the quality factor is 30, the BA of the proposed method is 3.47% lower than that of Stega_64. However, as the quality factor increases, the proposed method maintains the highest BA from a quality factor of 40 onward. Under extreme distortion scenarios, such as a crop ratio of 70%, the BA of all methods drops significantly, while the proposed method still attains the highest BA.

4. ABLATION STUDY

4.1. Effectiveness of HECAM

The effectiveness of HECAM is evaluated by comparing the proposed model with variants incorporating SE [15] and PSA [16] modules, as well as a baseline model without any attention mechanism. As shown in Table 5, the model with HECAM outperforms the other three ablation models in terms of invisibility and robustness, with improvements of 1.64 dB in PSNR and 1.95% in average BA compared to the model without HECAM. Even without HECAM, the WNet maintains BA over 95.5% across diverse shooting angles, indicating that long-range model can provide a strong baseline for screen-shooting robustness, while HECAM further boosts performance by enhancing texture-related high-frequency carriers. The effectiveness of HECAM lies in its high-frequency decomposition based on the Laplacian pyramid, which enables the network to capture global features while preserving local details.

4.2. Effectiveness of texture edge guided MSE loss

To evaluate the effectiveness of the proposed loss function in guiding the distribution of watermark features, L_e is replaced by the MSELoss in an ablation experiment. As shown in Fig.

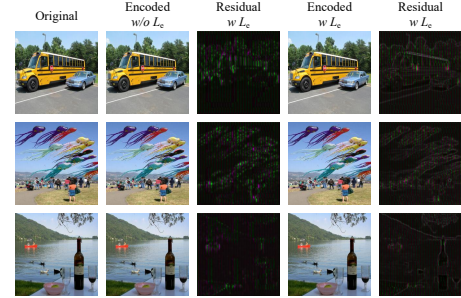


Fig. 3. Examples of watermarked images generated by the model with L_e and the model without L_e , along with their corresponding residual images (magnified 5 times).

Table 5. Comparison of models with various attention modules

		w HECAM	PSA	SE	w/o HECAM
Visual quality	PSNR (dB)	35.29	33.46	33.51	33.65
	SSIM	0.9868	0.9849	0.9851	0.9790
Attack	30 cm	99.0%	98.3%	98.2%	97.7%
	40 cm	98.5%	98.0%	97.8%	97.4%
	50 cm	98.6%	97.9%	98.0%	96.8%
	60 cm	98.2%	97.8%	97.8%	97.0%
	Left 30°	98.6%	97.4%	97.8%	96.3%
	Shooting Right 30°	99.0%	97.7%	98.0%	96.4%
	angle Up 30°	98.6%	98.0%	97.6%	95.9%
	Down 30°	98.3%	97.5%	97.3%	95.5%

3, the residual image generated by the model with L_e has a concentrated distribution of watermark information. This is because L_e encourages I_r to match the LDC-predicted mask, resulting in a more concentrated residual distribution on texture edge regions.

5. CONCLUSION

This work introduces a resilient watermarking scheme that significantly improves both robustness and imperceptibility against screen-shooting distortions. By employing long-range feature modeling and high-frequency enhancement techniques, the proposed method improves watermark localization accuracy and achieves good preservation of image details.

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#621 : "Screen-Shooting Resilient Watermarking based on Long-Range Modeling and High-Frequency Enhancement"

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Topics/EDICS:

Machine Learning for Signal Processing - Deep learning : Deep learning models

Image, Video and Multidimensional Signal Processing - Image and video processing techniques : Machine learning for image and video processing

Information Forensics and Security - Watermarking and data hiding : Theoretical models

Keywords:

Screen-shooting, Watermarking, Long-range modeling, High-frequency enhancement, Texture edge guided loss

REVIEW #1

Confidence: Confident

Recommendation: Marginal accept, Not award quality (score: 4.42)

Reviewer's comments to authors: |---

The authors introduce a useful and robust approach to screen-shooting resilient watermarking, demonstrating substantial technical novelty through the WNet cascaded dual-UNet architecture and the High-Frequency Enhancement Channel Attention Module (HECAM). The experimental depth is sufficient, effectively validating the model's resilience across various real-world shooting distances and angles on the COCO dataset. Furthermore, the literature review is well-constructed, and the references are relevant, providing excellent context regarding recent deep learning-based watermarking frameworks without any significant omissions. Overall, the clear structural progression and thorough empirical evidence make this a publication-ready paper.

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Confidential comments to chairs: |---

It's a good paper. The work has a moderate level of novelty and is useful.

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Is the work within the scope of the conference and relevant to ICIP? Clearly within scope

Is the technical contribution novel? Substantial novelty

Is the technical contribution significant? Moderate contribution

Is the manuscript technically correct? Technically sound

Is the manuscript properly structured and clearly written? Well-structured and clearly written

Is the level of experimental validation sufficient? Sufficient validation/theoretical paper

Are the references appropriate, without any significant omissions? No significant omissions

Are there any references that do not appear to be relevant? All references are directly relevant to the contribution of the manuscript

Is this paper ready to be extended to a full-length journal submission? Yes, ready to be extended to a full-length journal submission.

Would you be willing to serve as reviewer of the extended journal submission? No

REVIEW #2

Confidence: Very Confident

Recommendation: Marginal accept, Not award quality (score: 3.81)

Reviewer's comments to authors: |---

This paper proposes to develop a neural watermarking scheme which is robust to screen-shooting. The synchronisation process uses the 4 corner of the image. On originality is that is use a distortion layer, PIMoG, even if a comparison with basic data-augmentation processes could have been done. The performances seems good, but comparisons with PixelSeal or TrustMark, which are very robust to geometrical transform, are missing. The experimental setting is interesting and rigorous though. The security of this scheme is probably very small since the structure of the watermark is very simplistic (vertical waves, as seen in Fig. 3).

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Confidential comments to chairs: |---

Ok for ICIP in my opinion.

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Is the work within the scope of the conference and relevant to ICIP? Clearly within scope

Is the technical contribution novel? Substantial novelty

Is the technical contribution significant? Substantial contribution

Is the manuscript technically correct? Some minor concerns

Is the manuscript properly structured and clearly written? Some minor issues

Is the level of experimental validation sufficient? Limited but convincing

Are the references appropriate, without any significant omissions? Some significant omissions

Are there any references that do not appear to be relevant? Some of the references are of limited relevance

Is this paper ready to be extended to a full-length journal submission? No, not ready

Would you be willing to serve as reviewer of the extended journal submission? No

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REVIEW #3

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Confidence: Confident

Recommendation: Marginal accept, Not award quality (score: 4.42)

Reviewer's comments to authors: |---

This paper builds on watermark channel estimation for resisting screen-shooting (D/A/D conversion) to propose an architecture that:

1. uses skip connections to build a wavelet-type filter bank with down-/up-sampling capability (seeking clearer separation between analysis scales);

2. uses attention-based Laplacian enhancement of textures (where most watermark energy is usually carried).

The experimental part shows clear improvement, although maybe not as significant as the Authors claim, compared to the SOTA using neural nets. The respective merits of the two modules are assessed using sound ablation studies. In my opinion, the improvement is better overall robustness on more attacks, not so much the 0.52dB reduction in the embedding distortion.

I'd suggest only minor editing:

- I guess the word 'wavelet' has to appear somewhere, as the resulting architecture clearly builds on the same structure;

- I don't see any reason why the venerable Bit-Error-Rate (BER) should be replaced by the 'Bit Accuracy' that the authors promote, as $BER = 1 - BA$ and watermarking is fundamentally a variation on communications with side information.

Coming back to the results, what they suggest is:

- better modelling capability of the network when it can learn the wavelet filters (this is what these approaches change fundamentally compared to the classical perfect-reconstruction, bi-orthogonal framework);

- it is still unclear whether these techniques outperform the results obtained, e.g., by Deguillaume and Pun (SPIE, 2002) with the classical, much more energy-friendly and distortion-controlled schemes dating back 20+ years ago (that can embed 64 bits too, but at 38dB and resisting roughly the level and variety of attacks considered by the Authors).

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Confidential comments to chairs: |---

See review.

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Is the work within the scope of the conference and relevant to ICIP? Clearly within scope

Is the technical contribution novel? Substantial novelty

Is the technical contribution significant? Moderate contribution

Is the manuscript technically correct? Technically sound

Is the manuscript properly structured and clearly written? Well-structured and clearly written

Is the level of experimental validation sufficient? Sufficient validation/theoretical paper

Are the references appropriate, without any significant omissions? No significant omissions

Are there any references that do not appear to be relevant? All references are directly relevant to the contribution of the manuscript

Is this paper ready to be extended to a full-length journal submission? No, not ready

Would you be willing to serve as reviewer of the extended journal submission? No

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AUTHORS' REBUTTAL:

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Response to Reviewer 1

We thank the reviewer for the positive assessment.

Response to Reviewer 2

2.1 Role of PIMoG and comparison with PixelSeal/TrustMark

In our work, PIMoG is not the main novelty but a training component used to approximate the compound degradations of screen-shooting. The main contribution is the cascaded dual-UNet-based watermarking framework and the high-frequency enhancement channel attention module.

PixelSeal and TrustMark are important references for conventional geometric and electronic-channel distortions, but such robustness does not transfer to screen-shooting, which also introduces illumination changes, blur, moire, and device artifacts. Our reproduction shows that PixelSeal outperforms TrustMark. In its original setting, PixelSeal achieves 93.8%, 95.0%, and 96.3% under JPEG-50, Cropout-0.3, and resize-0.4, with a PSNR of 36.75 dB. However, when directly evaluated under the screen-shooting scenario, its performance is limited. At a shooting angle of 20° and a shooting distance of 50 cm, its decoding BA is only 84.3%. After retraining it under the same resolution, payload, and PIMoG-based noise layer as our method, its BA under screen-shooting improved to 95.8% at the same shooting setting, but its PSNR decreased to 34.19 dB, and its Cropout-0.3 robustness also declined markedly. This indicates a trade-off between screen-shooting robustness, visual quality, and conventional-channel robustness. Our method shows more stable performance across shooting angles, distances, and devices.

2.2 Security implication of Fig. 3

Fig. 3 illustrates the embedding structure rather than a directly observable fixed watermark template. Message recovery relies on learned decoding, not direct reading of a visible pattern. We will clarify that our work focuses on robustness under screen-shooting conditions, while cryptographic security analysis is outside its scope.

Response to Reviewer 3

3.1 Relation between WNet and wavelet-type filter bank structures

We thank the reviewer for this point. WNet shares a multi-scale analysis-reconstruction characteristic with wavelet-type filter bank frameworks. However, the method does not use a classical wavelet transform. The analysis and fusion processes are learned through convolution. The backbone is a learnable cascaded dual-UNet with two-stage encoding/reconstruction and dense cross-stage skip connections. The relation is structural rather than based on a wavelet-filter design.

3.2 BA and BER

We used BA for consistency with recent neural watermarking papers and because it directly reflects the proportion of correctly decoded bits. The two metrics are equivalent: $BER = 1 - BA$. We will make this relation explicit in the revision.

3.3 Discussion of Deguillaume and Pun (SPIE, 2002)

Deguillaume and Pun is a representative classical method with strong affine synchronization capability. However, our work focuses on the screen-shooting setting rather than general affine robustness. Since the task assumptions and evaluation protocols are not fully aligned, direct comparison would not be rigorous. We will narrow the claim to 'under the considered screen-shooting scenario'.

3.4 Discussion of traditional watermarking methods

Following the reviewer's suggestion, we examined traditional methods represented by Pun et al. However, some implementation details could not be recovered reliably from the public descriptions, so we could not obtain stable results for a fair comparison. We also examined classical transform-domain baselines such as the DWT-DCT-SVD family. Although these methods are effective under standard digital distortions, their robustness can not transfer reliably to the screen-shooting protocol. This is because screen-shooting involves the display-camera pipeline, which alters photometric response and sampling structure. In this case, traditional schemes designed primarily for conventional digital-channel distortions may have limited robustness in a screen-shooting scenario.

(no chair comments to authors)